



THE BEHAVIOR OF 6G SYSTEM FOR DIFFERENT BANDWIDTHS

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ABSTRACT

It is a 6G technology that is proposed to be upgraded to 5G technology. It is expected to integrate 5G with satellite networks for global coverage. 6G technology is considered to be cheap and fast, providing high data rates and fast Internet speed to access on air through wireless and mobile devices with data ranges up to 11 Gbps. This report examines the relationship between spectral efficiency and Signal to Interference plus Noise Ratio (SINR) in the context of sub-Tera Hertz system.

INTRODUCTION

1. Evolution of Wireless Technologies

Wireless communication has come a long way from **1G to 5G**, with each generation introducing significant advancements in speed, connectivity, and application capabilities

Ex: Wireless symbol with progressive arrows

2. The Need for Faster, More Reliable Networks

With the exponential rise in data traffic, IoT devices, and AI integration, the need for and more dependable networks is pushing the demand for 6G

Ex: Speedometer with high-speed mark

3. What is 6G?

6G is the next major leap in **wireless communication**, promising faster speeds, lower latency, and global connectivity. It will unlock new possibilities for both individuals and industries.

Ex: Satellite network globe

4. 6G Beyond Mobile Connectivity

While 5G focuses on mobile broadband, 6G aims to extend connectivity to autonomous systems, smart cities, and **holographic communication**.

Ex: Holographic projection

5. Global Coverage with Satellite Integration

Unlike 5G, which depends on terrestrial infrastructure, 6G integrates satellite networks for seamless global coverage, ensuring connectivity in even the most remote regions.

Ex: Satellite orbiting Earth



Literature Survey

Ref . No	Title of the Pages	Findings
1	"6G Wireless Networks: Vision, Requirements, and Key Technologies"	50-100 GHz enables high data rates, but signal loss and line-of-sight limitations impact range. Power, antenna gains, and interference management are crucial for performance.
2	"Terahertz Communication for 6G Networks"	150-200 GHz offers ultra-high speeds but suffers from high signal loss and short range. Dense infrastructure is needed to overcome these challenges.
3	"Opportunities and Challenges of Sub-THz Communication in 6G"	100-150 GHz provides high throughput for IoT but faces challenges with interference and path loss. Optimizing gains and minimizing noise is essential for efficiency.
4	"Millimeter-Wave and Sub-Terahertz Band for 6G: Spectrum and Propagation Characteristics"	50-100 GHz offers high-speed communication but is affected by high attenuation, requiring careful management of noise and interference.
5	"Channel Modeling and Propagation for Terahertz Communication in 6G"	150-200 GHz supports terabit-per-second speeds but is limited by severe path loss, requiring advanced antenna designs and power control.

Ref . No	Title of the Pages	Findings
6	"Towards 6G:The Role of the 10-20 MHz Frequency Band in Low-Power IoT Applications"	10-20 MHz is ideal for long-range, low-power IoT communication with minimal signal loss, but provides lower data speeds.
7	"The Impact of Frequency Bands on 6G System Design"	Higher frequencies like 100-150 GHz offer fast speeds but suffer from high interference and path loss. Lower frequencies like 10-20 MHz offer better coverage at slower speeds.
8	"Challenges and Opportunities of 6G Wireless Communications:A Survey"	100-150 GHz is suitable for smart cities, but interference and noise must be minimized for optimal performance.
9	"Advanced Antennas and Propagation for Terahertz Wireless Systems"	150-200 GHz offers high speeds but requires advanced antenna designs to overcome severe signal attenuation and short-range limitations.
10	"Low-Power Communication in 6G:The Role of the 10-20 MHz Band"	10-20 MHz is ideal for long-range IoT communication with minimal signal loss, although it offers slower speeds compared to higher frequencies.

Theory

- 50 GHz to 100 GHz Band:**

- Benefits:** Offers fast data rates and low latency.
- Applications:** Great for VR, AR, and URLLC (Ultra-Reliable Low-Latency Communication).
- Challenges:** Shorter range due to higher signal loss and need for line of sight.

- 100 GHz to 150 GHz Band:**

- Benefits:** Provides even faster data speeds.
- Applications:** Ideal for smart cities and advanced IoT.
- Challenges:** Best for short-range, high-capacity uses, with higher signal loss and weather sensitivity.

- 150 GHz to 200 GHz Band:**

- Benefits:** Can achieve data speeds in terabits per second.
- Applications:** Perfect for high-precision technologies like AI and holography, and huge data transfers.
- Challenges:** Requires expensive infrastructure and has very short range due to high signal loss.

- 10 MHz to 20 MHz Band:**

- Benefits:** Provides wide coverage and less signal loss.
- Applications:** Suitable for long-range communication and IoT devices.
- Challenges:** Slower speeds and limited bandwidth due to current use in mobile networks.

- Summary:**

- Lower frequencies:** Better for coverage but slower speeds.
- Higher frequencies:** Offer faster speeds but shorter range.



Table-1: 5G vs 6G

Aspect	5G	6G
Generation	Fifth Generation (5G)	Sixth Generation (6G)
When it Arrived	Started rolling out in 2019-2020	Expected around 2030
Speed	Super fast, up to 10 Gbps	Superfast fast, up to 1 Tbps
Delay (Latency)	Very low, ~ 1ms	Even lower(~0.1 ms)
Frequency	Approx 24–100 GHz	Approx 100 GHz–10 THz
Bandwidth	Moderate (up to 400 MHz)	Huge, several times broader than 5G
Devices Connected	Handles 1 million devices per square Km	Connect 10 million devices in same area
Energy Use	Energy-efficient compared to 4G	Even greater energy efficiency with AI
Main Uses	Smart cities, IoT, automation	Holograms,AR/VR, space networks
Cool Tech Behind It	Massive MIMO, Beamforming, Network Slicing	Powered by AI, quantum communication
Security	Safer than older systems	Ultra-secure with quantum-proof encryption



Shannon Hartley's Channel Capacity

Enhanced Capacity Metrics :

From Shannon Hartley's Theorem, channel capacity C may be given as:

$$C = B * \log_2(1 + SNR) \dots\dots\dots[1]$$

Where B: Bandwidth of communication channel

SNR: Signal to Noise Ratio



SUB-TERA HERTZ SYSTEM (Freq Range= 100 to 300 GHz)

The SINR of the this 6G system is defined as:

$$S(y) = \frac{P_A G_A G_U S_F U_F}{(N_0 W + I) L(y)} \dots\dots\dots [2]$$

and Spectral efficiency $R = \log_2[1 + s(y)]$

Where,

S(y): SINR for user y

P_A: Transmitted Power

G_A: Antenna Gain at Base Station

G_U: Antenna Gain at user end

S_F: Shadow fading

U_F: User fraction

N₀: Noise power spectral density

W: Bandwidth of the channel

I: Interference

L(y): Channel loss



SPECTRAL EFFICIENCY FOR DIFFERENT BANDWIDTHS

Figure: I

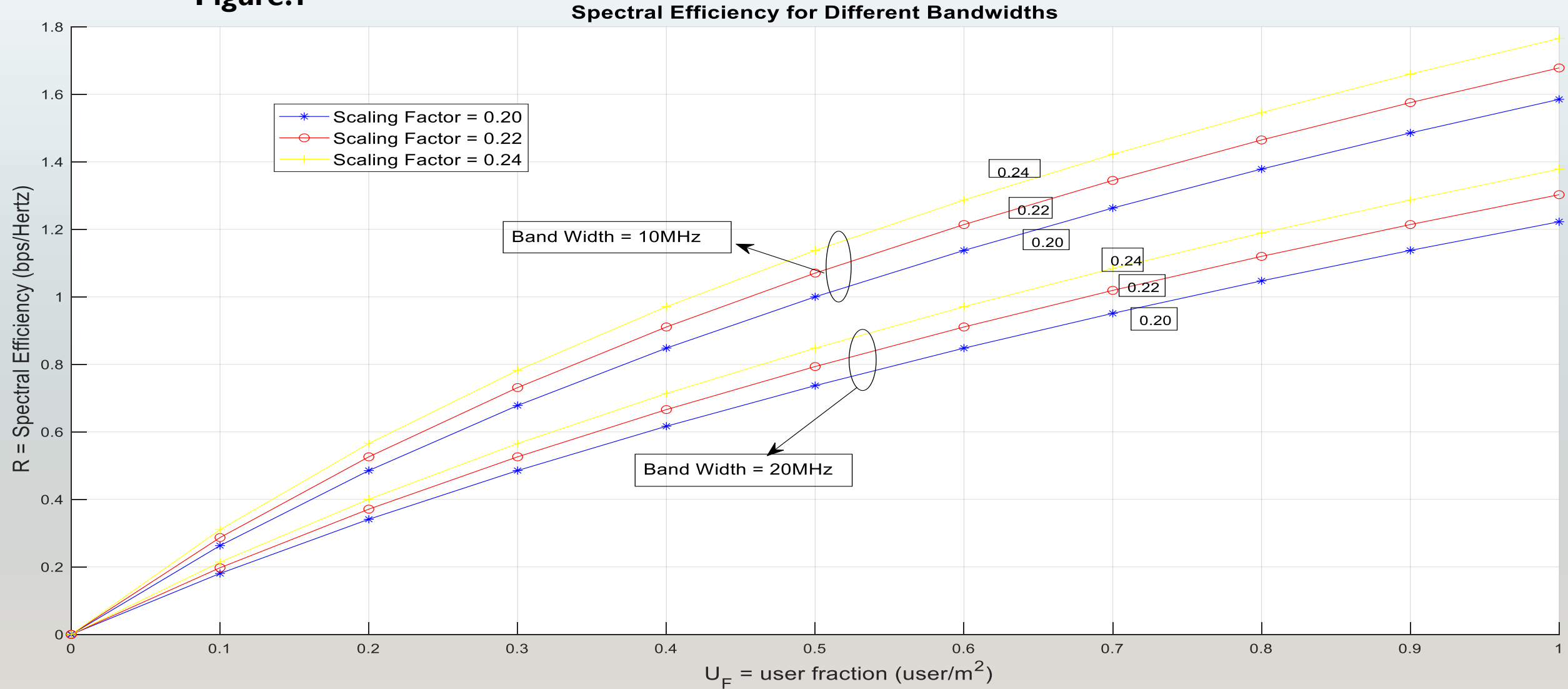


Fig. 1: The Figure shows the for constant Scaling Factors and constant User Fraction, if we increases Band Width then Spectral efficiency decreases, here lower ring have highest BW i.e. 20MHz but spectral efficiency is less as compared to that of 10MHz.

On the other hand, for constant Band Width & constant User Fraction, if we increase the Scaling Factors the Spectral Efficiency (R) increases, here yellow line has more Spectral Efficiency(R).

Figure:2

Spectral Efficiency for Different Bandwidths

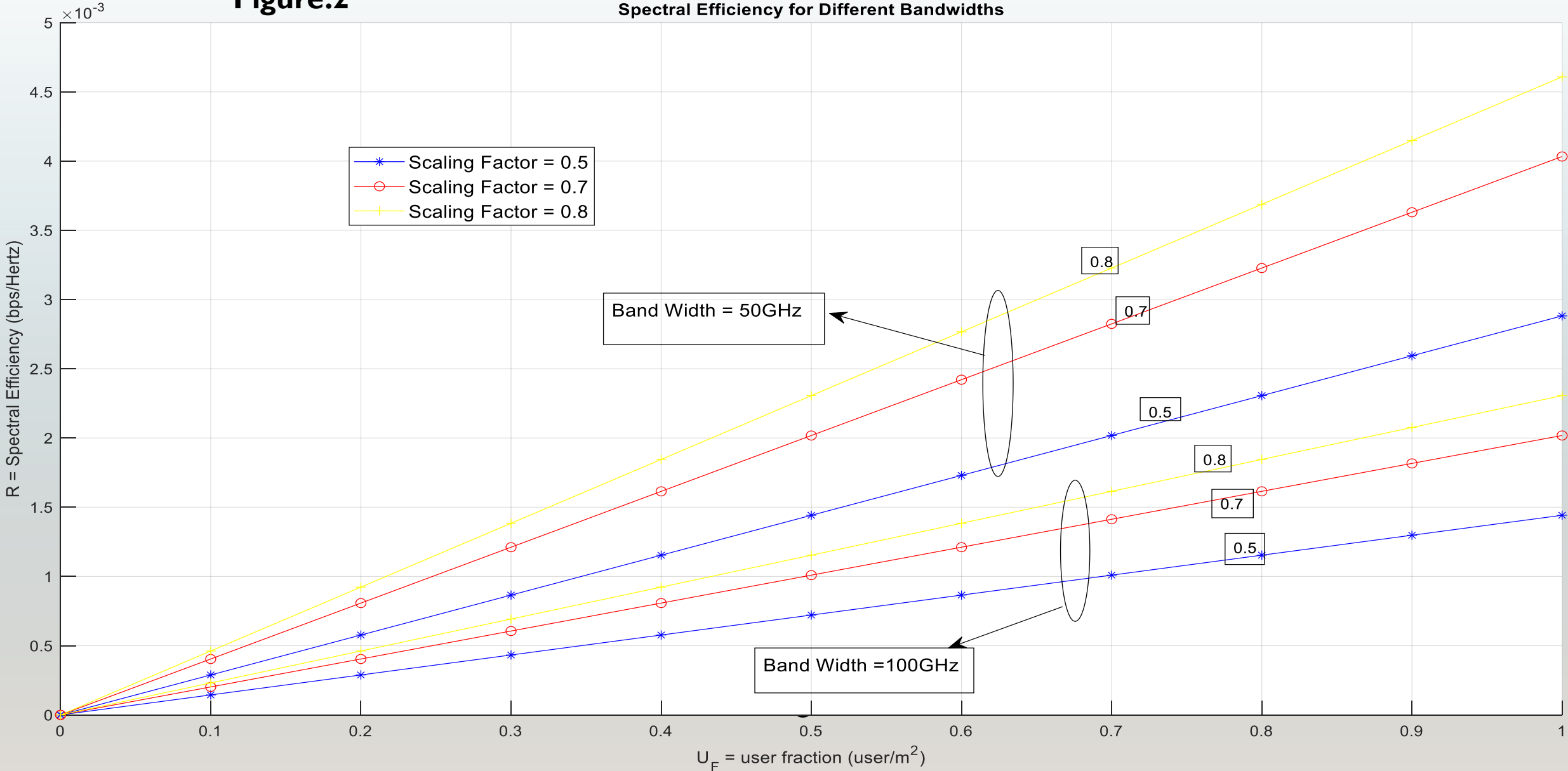


Fig. 2: The Figure shows the for constant Scaling Factors and constant User Fraction, if we increases Band Width then Spectral efficiency decreases, here lower ring have highest BW i.e. 100GHz but spectral efficiency is less as compared to that of 50GHz.

On the other hand, for constant Band Width & constant User Fraction, if we increase the Scaling Factors the Spectral Efficiency (R) increases, here yellow line has more Spectral Efficiency(R).

Figure:3

Spectral Efficiency for Different Bandwidths

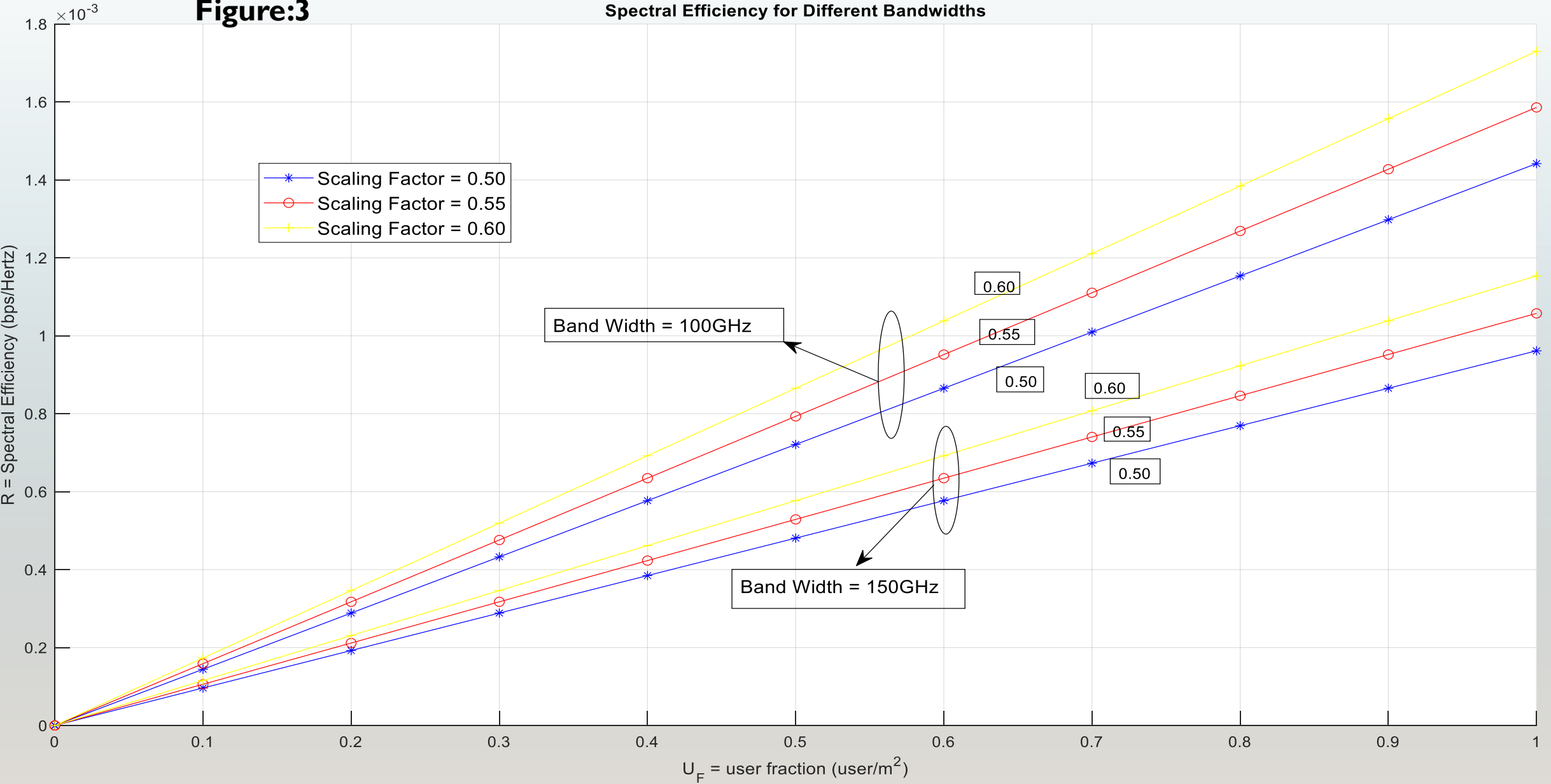


Fig. 3: The Figure shows the for constant Scaling Factors and constant User Fraction, if we increases Band Width then Spectral efficiency decreases, here lower ring have highest BW i.e. 150GHz but spectral efficiency is less as compared to that of 100GHz.

On the other hand, for constant Band Width & constant User Fraction, if we increase the Scaling Factors the Spectral Efficiency (R) increases, here yellow line has more Spectral Efficiency(R).

Figure:4

Spectral Efficiency for Different Bandwidths

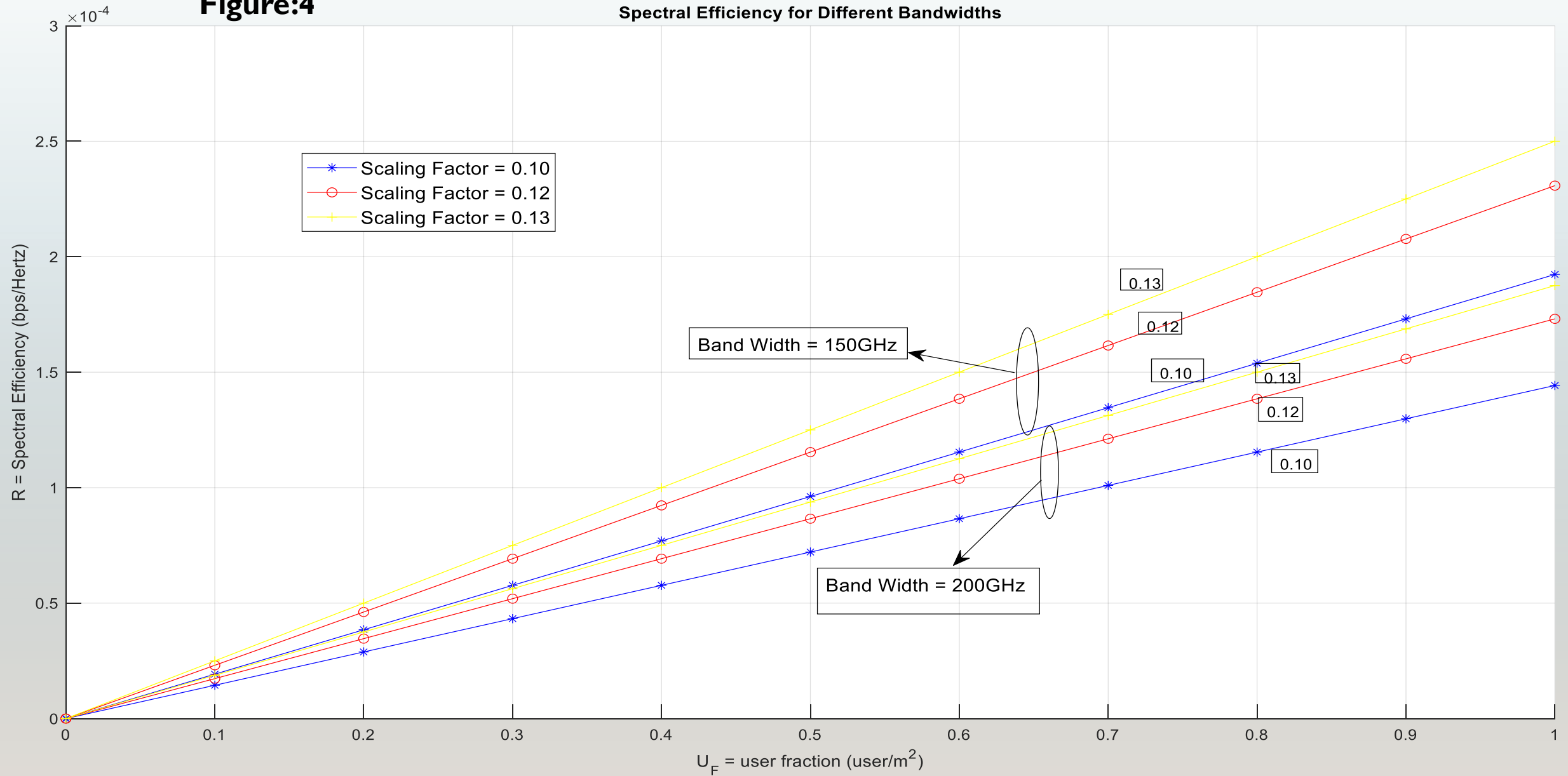


Fig. 4: The Figure shows the for constant Scaling Factors and constant User Fraction, if we increases Band Width then Spectral efficiency decreases, here lower ring have highest BW i.e. 200GHz but spectral efficiency is less as compared to that of 150GHz.

On the other hand, for constant Band Width & constant User Fraction, if we increase the Scaling Factors the Spectral Efficiency (R) increases, here yellow line has more Spectral Efficiency(R).

ANALYSIS & RESULTS:

System Parameters :

- For Band Widths 10MHz, 20MHz,
Scaling Factors are 0.20 (blue line) , 0.22 (red line) & 0.24 (yellow line)
- For Band Widths 50GHz, 100GHz,
Scaling Factors are 0.5 (blue line) , 0.7 (red line) & 0.8 (yellow line)
- For Band Widths 100GHz, 150GHz,
Scaling Factors are 0.50 (blue line) , 0.55 (red line) & 0.60 (yellow line)
- For Band Widths 150GHz, 200GHz,
Scaling Factors are 0.10 (blue line) , 0.12 (red line) & 0.13 (yellow line)

DISCUSSION

Impact of Scaling factor on Spectral Efficiency

- Scaling factors directly affect spectral efficiency (bits per second per Hz). Higher scaling factors allow more information to be transmitted within a given bandwidth.

Scaling Factor $\uparrow \Rightarrow R \uparrow$ (constant U_F , constant BW)

- According to the figure (graphs)

- **Fig. 1:** Bandwidths = 10 MHz and 20 MHz

Curves are plotted for scaling factors of **0.20, 0.22, and 0.24**.

At $U_F=0.5$, the yellow curve (0.24) achieves the highest spectral efficiency.

- Similarly, we can say for remaining three fig.

Impact of Band Width on Spectral Efficiency

BW $\uparrow \Rightarrow R \downarrow$

(constant U_F , constant Scaling Factor)

- Two bandwidths are considered: **10 MHz and 20 MHz (Fig. 1)**

The spectral efficiency decreases as bandwidth increases.

For example: At $U_F=0.5$, the 20 MHz curves are below the 10 MHz curves for all scaling factors

6G System Model for Different Bandwidths

- Frequency Bands:**

- GHz (50-200 GHz): **Fast, short-range** communication for **urban** areas.
- MHz (10-20 MHz): **Long-range**, dependable communication for **rural** areas.

- Scaling Factor:** Adjusts performance based on power, range, and signal travel; higher for GHz, lower for MHz .

- User Fraction:** GHz for dense areas, MHz for narrow areas.

- Spectral Efficiency:** GHz offers high data efficiency, MHz is simpler with lower efficiency.

- MATLAB Simulation:** Models how bandwidth, user distribution, and efficiency interact in different scenarios.

The image highlights the key features of 6G system ultra-high bandwidth, precision, performance, energy efficiency, massive connectivity, and intelligence. These characteristics aim to revolutionize wireless communication and enable a future with faster, more reliable, and intelligent networks.



CONCLUSION

In the final analysis, the MATLAB-plotted graphs for bandwidths ranging from 50-100 GHz, 100-150 GHz, 150-200 GHz, and 10-20 MHz prove how the 6G system behaves differently across various frequency bands. For constant User Fraction & constant Band Width, high Scaling Factor shows high Spectral Efficiency, but in other hand for constant User Fraction & constant Scaling Factor, high Band Width shows high Spectral Efficiency. So, for high Spectral efficiency shows high data rate. Sub-THz systems operate in the frequency range of 100-300GHz. 6G will make a revolution with ultra-fast speeds, sub-THz frequencies, AI driven intelligent networks, AR/VR, holography.



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THANK

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